

Osman Şahin Güler

Software Engineering Student

Summary

Sophomore Software Engineering student with a deep technical focus and hands-on experience in computer vision, autonomous systems, and embedded software development. Actively developing object detection pipelines using YOLO architectures and autonomous navigation algorithms via the ArduPilot ecosystem for Unmanned Aerial Vehicles (UAVs). Committed to transforming technical expertise into high-performance, scalable engineering solutions.

Education

Sept 2024 – Present **B.Sc. in Software Engineering**, *Yaşar University*, Izmir, GPA: 3.61 / 4.00
Expected Graduation: June 2027

Experience

Sept 2024 – Present **Autonomous Systems & Software Developer**, *Yaşar-Anafrata UAV Team*, Izmir

- **Computer Vision:** Optimized high-accuracy detection pipelines using Pinhole Camera models and YOLO (v8, v11) architectures for small object detection in high-altitude aerial imagery.
- **Autonomous Mission Management:** Established telemetry data processing systems via ArduPilot and MAVLink protocols; enabled autonomous takeoff, navigation, and target-oriented mission execution.
- **Simulation & Testing:** Conducted 50+ hours of real-time physical simulations and scenario testing in Gazebo and SITL (Software In The Loop) environments to validate algorithm safety and reliability.
- **Embedded System Optimization:** Implemented hardware acceleration techniques and low-latency video streaming using GStreamer on edge devices (Raspberry Pi).
- **Infrastructure & DevOps:** Integrated Docker-based microservices and Portainer management systems to ensure environmental consistency and streamline deployment processes.

Izmir, Turkey

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- Jan 2025 – **Software Developer (Intern/Prototype)**, *CLB Automation*,
Oct 2025 Remote/Hybrid
- Developed computer vision prototypes for industrial object detection using OpenCV and YOLOv5.
 - Enhanced detection accuracy through model training and hyperparameter optimization on custom-labeled datasets.
 - Conducted performance profiling and optimization on real-time data processing pipelines.

Technical Skills

Programming	Python (Proficient), C++, JavaScript (React)
AI & Vision	OpenCV, YOLO (v5, v8, v11), SAHI, NumPy, Data Processing
Robotics	ArduPilot, MAVLink, Gazebo, Mission Planner, SITL
DevOps & Tools	Docker, Portainer, Git, Linux (Ubuntu), Raspberry Pi
Engineering	Data Structures & Algorithms, Object-Oriented Programming (OOP)

Languages

English	Fluent	<i>Professional working proficiency</i>
Japanese	Beginner	<i>Basic communication</i>

Interests & Achievements

Sports	Sailing (10 Years Professional Licensed Athlete - Teamwork & Leadership)
Martial Arts	Regular training focused on discipline, focus, and physical endurance